

# YCQE-EHT-QF-13.1 EHT INSTALATION/ INSULATION TEST REPORT

|                     |              |                      |                      |                   |         |
|---------------------|--------------|----------------------|----------------------|-------------------|---------|
| <b>CLIENT NAME:</b> | COOEC Canada | <b>PROJECT NAME:</b> | K1B WELL PAD PROJECT | <b>CONTACT #:</b> | CA23007 |
|---------------------|--------------|----------------------|----------------------|-------------------|---------|

## Heat trace information

|                      |                                            |                 |               |                 |                                                                             |
|----------------------|--------------------------------------------|-----------------|---------------|-----------------|-----------------------------------------------------------------------------|
| <b>TRACE PART #:</b> | D/32SP4734/90M/3006/277/1.2M/LS23A/X/N12   |                 |               |                 |                                                                             |
| <b>TRACE #:</b>      | 29103-ET-PG1101-1-1                        |                 |               | <b>LENGTH:</b>  | 90 m                                                                        |
| <b>LINE NUMBER:</b>  | 29103-PG-1101, 29103-PE-1102, 29103-P-1100 |                 |               | <b>WATTS/M:</b> | 33.4 W/m                                                                    |
| <b>DRAWING #:</b>    | 29103-65R-ET-PG1101-01-01                  |                 |               | <b>TYPE:</b>    | <input type="checkbox"/> SELF REG. <input checked="" type="checkbox"/> M.I. |
| <b>CONTROLLER #:</b> | HTC-15                                     | <b>PANEL #:</b> | 29103-HTP-002 | <b>CIRCUIT:</b> | CB-15                                                                       |

## Test equipment data

|                                |                       |  |               |  |                 |  |
|--------------------------------|-----------------------|--|---------------|--|-----------------|--|
| <b>MULTI-METER INFORMATION</b> | <b>SERIAL NUMBER:</b> |  | <b>MODEL:</b> |  | <b>CAL DUE:</b> |  |
| <b>MEGGER</b>                  | <b>SERIAL NUMBER:</b> |  | <b>MODEL:</b> |  | <b>CAL DUE:</b> |  |

## Megger test voltage as per data sheet

|                             |                                  |                                   |                                   |               |  |
|-----------------------------|----------------------------------|-----------------------------------|-----------------------------------|---------------|--|
| <b>MEGGER TEST VOLTAGE:</b> | <input type="checkbox"/> 500 VDC | <input type="checkbox"/> 1000 VDC | <input type="checkbox"/> 2500 VDC | <b>OTHER:</b> |  |
|-----------------------------|----------------------------------|-----------------------------------|-----------------------------------|---------------|--|

### 1. RECEIVED EHT MEGGER

#### TESTING CONDITIONS

| INSULATION M/G $\Omega$ | CONTINUITY $\Omega$ | STATUS | INITIAL | DATE-Year-M-D | TEMP. °C | HUMIDITY |  |
|-------------------------|---------------------|--------|---------|---------------|----------|----------|--|
|                         |                     |        |         |               |          |          |  |

### 2. AFTER INSTALLATION MEGGER

#### TESTING CONDITIONS

| INSULATION M/G $\Omega$ | CONTINUITY $\Omega$ | STATUS | INITIAL | DATE | TEMP. °C | HUMIDITY |  |
|-------------------------|---------------------|--------|---------|------|----------|----------|--|
|                         |                     |        |         |      |          |          |  |

### 3. AFTER INSULATION MEGGER

#### TESTING CONDITIONS

| INSULATION M/G $\Omega$ | CONTINUITY $\Omega$ | STATUS | INITIAL | DATE | TEMP. °C | HUMIDITY |  |
|-------------------------|---------------------|--------|---------|------|----------|----------|--|
|                         |                     |        |         |      |          |          |  |

### 4.Final commissioning (when applicable)

|                           |                 |  |                       |  |                        |  |
|---------------------------|-----------------|--|-----------------------|--|------------------------|--|
| <b>ENERGIZED RESULTS:</b> | <b>VOLTAGE:</b> |  | <b>START-UP AMPS:</b> |  | <b>OPERATING AMPS:</b> |  |
|---------------------------|-----------------|--|-----------------------|--|------------------------|--|

## Notes

**PART 5 – Visual Inspection**

| No. | Visual inspection requirement                                                                                                                                                                                                                                               | Initial / N/A |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1   | Upon receiving, but prior to installation, perform a continuity check and an Insulation Resistance Test from conductor to sheath (for MI cable); conductor to braid (for SR cable). Document on Test Record Form YCQE-E&I-002-Rev.0.                                        |               |
| 2   | Cables protected from severe blows and welding slag.                                                                                                                                                                                                                        |               |
| 3   | The correct cable is on the correct pipe and in the correct location. Record the line numbers above. Ensure correct EHT allowances for valves, shoes, drains etc. have been provided per details and drawing. Ensure constant wattage cables are 50mm away from each other. |               |
| 4   | Heat transfer cement or aluminum tape applies as indicated in the drawings and specifications.                                                                                                                                                                              |               |
| 5   | Perform continuity and insulation Resistance Tests after cable installation on pipe. Document tests in section 2 of this form.                                                                                                                                              |               |
| 6   | Before releasing for insulation, check that heating cable is properly strapped to pipes (adhesive tape for SR cable and SS wire for MI cable).                                                                                                                              |               |
| 7   | Check location of power connection points, RTDs, splice points, and end seals (End Kits), etc. with the drawings. Document all changes on applicable ISO.                                                                                                                   |               |
| 8   | Check power connection points, RTDs, trace tags have the proper identification as per drawing.                                                                                                                                                                              |               |
| 9   | MI cable power JB's installed so as not to interfere with insulation. (N/A for SR cables).                                                                                                                                                                                  |               |
| 10  | Perform continuity and Insulation Resistance Tests after insulation has been installed on pipe. Document tests on in part 5 of this document.                                                                                                                               |               |
| 11  | Check EHT warning labels have been placed on pipe insulation jacket at spacing specified in job specifications.                                                                                                                                                             |               |
| 12  | Confirm correct SR type is installed as per ISO, Wattage per ft/M.                                                                                                                                                                                                          |               |

**Notes**

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**Inspected/Approved By**

| Yanda QA Representative | Client Representative |
|-------------------------|-----------------------|
| Name:                   | Name: Levy Ostrup     |
| Date:                   | Date:                 |
| Signature:              | Signature:            |